

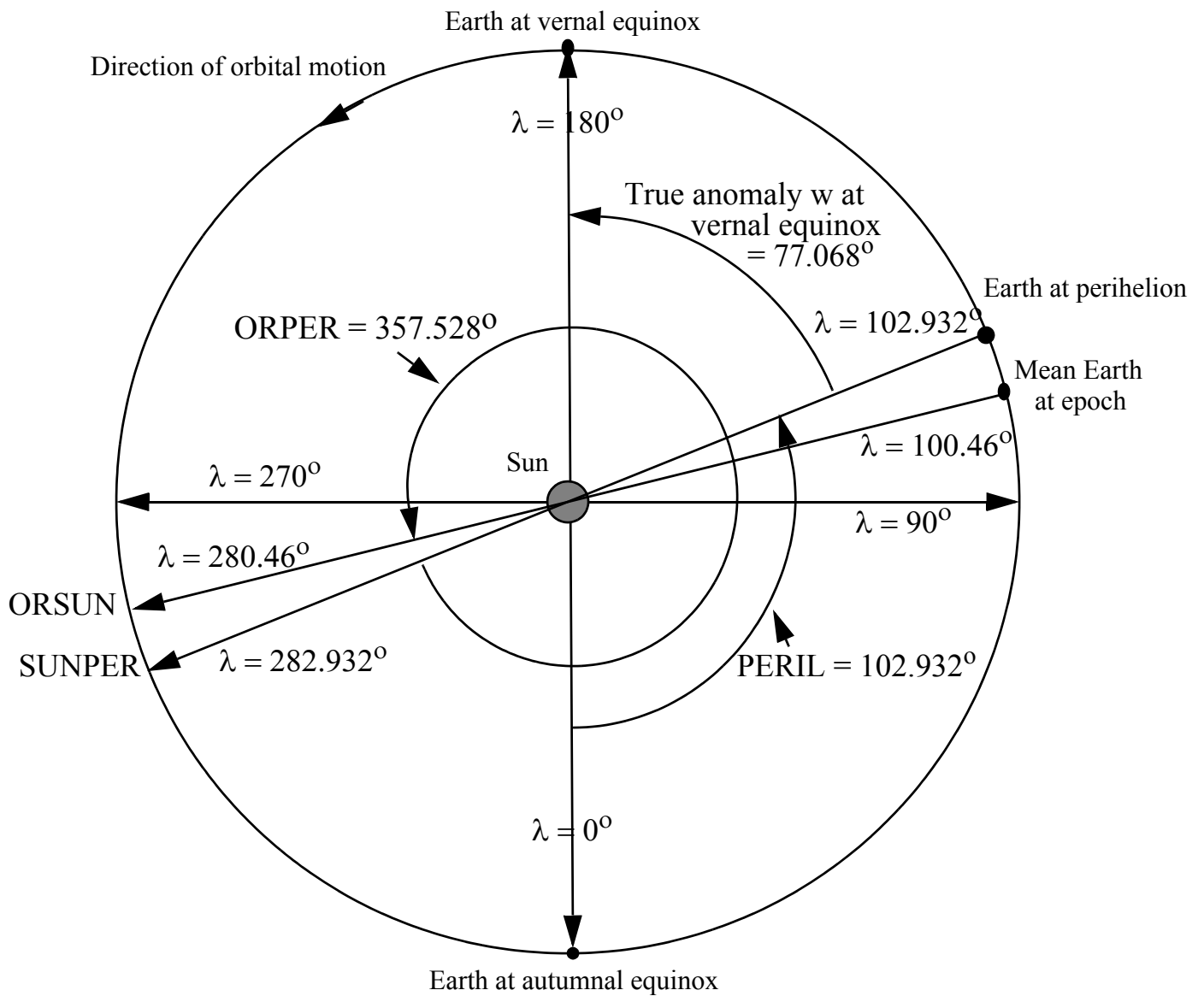


Figure A-1. Plan view of the Earth's orbit from the north showing the parameters described in the text. The angle is the celestial (ecliptic) longitude with an origin of zero degrees in the direction of the Sun at the vernal equinox. Values of the parameters (denoted by upper case letters) are those at the standard epoch (J2000.0). The orbit is drawn as a circle for simplicity, and some of the angles between position vectors have been exaggerated for clarity.

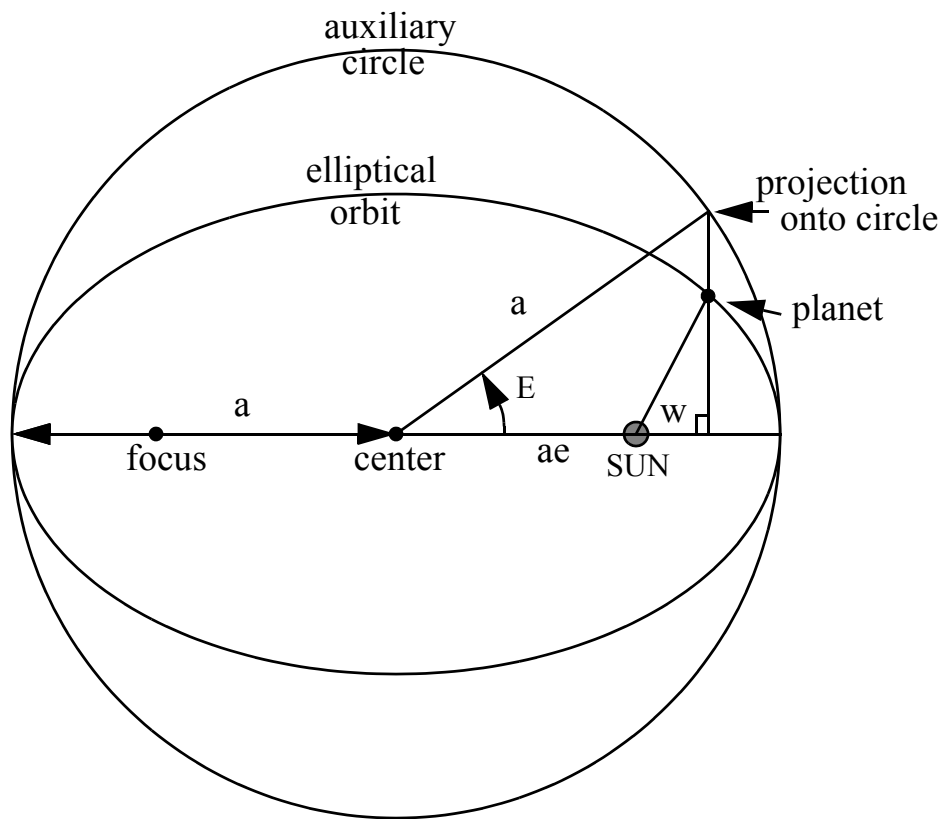


Figure A-2. An elliptical orbit circumscribed by an auxiliary circle drawn to illustrate the eccentric anomaly E and the true anomaly w . The Sun is at one focus of the ellipse, and the other focus is located symmetrically on the left side of the ellipse. The semimajor axis of the ellipse is a and half of the distance between the foci is ae , where e is the eccentricity of the ellipse.

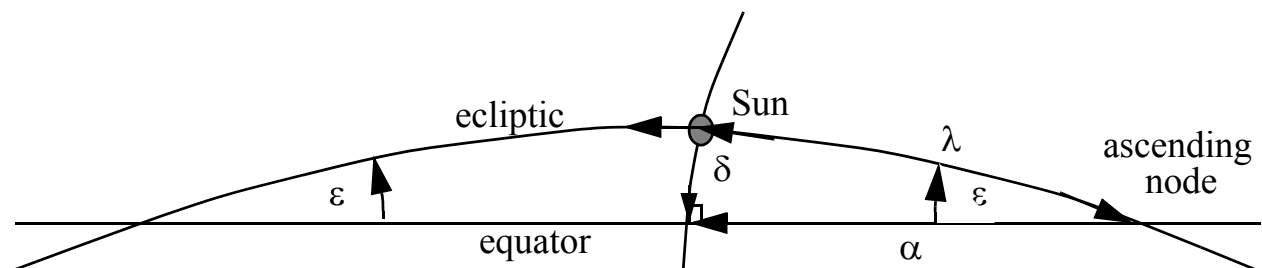


Figure A-3. The right triangle bounded by three great circles, i. e., the ecliptic, the equator and the hour circle passing through the Sun. If the obliquity ϵ and the ecliptic longitude of the Sun λ in this triangle are known, the solar declination δ and right ascension α can be determined by spherical trigonometry.